

5. Salesforce Development

5.1 Custom Application

A custom Lightning application named **Student Management System** was developed using Salesforce App Manager to provide a centralized platform for managing student information. The application contains custom tabs, reports, dashboards, and objects related to student management. It offers an organized interface for administrators and faculty to access all CRM functionalities from a single location.

5.2 Custom Objects

A custom object named **Student** was created to store and manage student information within Salesforce. The object serves as the primary entity of the CRM and maintains records such as Student ID, Department, Year, Email, and other academic details. This object allows users to efficiently create, update, and retrieve student records.

5.3 Custom Fields

Several custom fields were added to the Student object to capture essential student information. These fields include Student ID, Department, Year, Email, Phone Number, and Status. Appropriate data types such as Text, Picklist, Number, and Email were used to ensure accurate data entry and validation.

5.4 Page Layout

The Student page layout was customized to provide an intuitive user interface. Frequently used fields were arranged logically to improve usability and simplify record creation. The customized layout ensures that users can quickly access and update important student information.

5.5 List Views

A custom list view named **Final Year Students** was created to display only students whose academic year is four. This feature enables administrators to quickly identify final-year students for placement activities, academic reviews, and communication.

5.6 Reports

A report named **Student Details** was created to generate structured information about student records. The report includes important columns such as Student Name, Department, Year, and Email, enabling users to analyze and monitor student data efficiently.

5.7 Dashboards

A dashboard named **Student Dashboard** was developed to provide graphical insights into student information. Charts such as Students by Department and Students by Academic Year allow administrators to understand student distribution and make informed decisions based on visual analytics.

5.8 Validation Rules

Validation rules will be implemented to ensure data accuracy and consistency. These rules will prevent users from entering invalid or incomplete information, such as incorrect email formats or duplicate student IDs. Validation enhances the reliability and integrity of the CRM database.

5.9 Flow Automation

Record-Triggered Flows will be created to automate repetitive tasks within the CRM. For example, a flow can automatically assign a student status or send notifications when new student records are created. Automation reduces manual effort and improves business efficiency.

5.10 Approval Process

An approval process will be designed to automate the verification of student registration requests. Submitted records will pass through an approval hierarchy before becoming active, ensuring proper authorization and compliance with institutional policies.

5.11 Apex Classes and Triggers

Custom Apex classes and triggers will be developed to implement business logic that cannot be achieved through declarative tools. Triggers will validate data before records are saved, while Apex classes will perform advanced processing and automation.

5.12 Security Model

Salesforce security features such as Profiles, Roles, Permission Sets, Sharing Rules, and Field-Level Security will be configured to ensure that users access only the data relevant to their responsibilities. This approach maintains data confidentiality and supports secure CRM operations.

5.13 Reports and Dashboards

Custom reports and dashboards provide real-time visibility into student records and institutional performance. These analytical tools assist administrators in monitoring student enrollment, department distribution, and academic statistics through interactive visualizations.

5.14 Data Migration

The Data Import Wizard will be used to import student records from external sources into Salesforce. This process minimizes manual data entry and ensures efficient migration of large datasets while maintaining data integrity.

5.15 Testing

The CRM application will undergo comprehensive testing to verify the functionality of objects, validation rules, flows, reports, dashboards, and security settings. Test cases will be prepared with sample input and expected output to ensure that every feature performs according to business requirements.